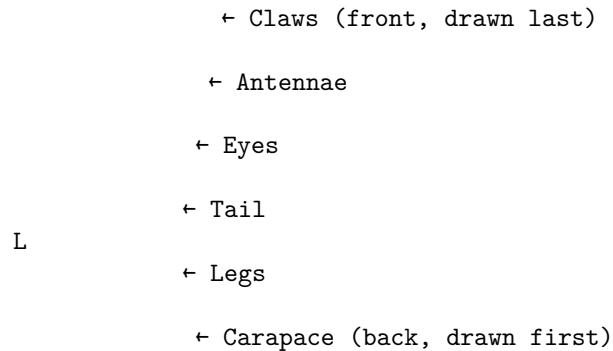


Composite Mode — Designer Guide

How Clawbada Lobsters Are Built

Every lobster in Clawbada is assembled from **6 body part layers** that are **stacked on top of each other** in the same 48x48 pixel space. Think of it like 6 transparent sheets of acetate laid over each other — each sheet has one part of the lobster drawn on it.



All 6 parts share the SAME canvas space.
They overlap - they are NOT side by side.

This is the key concept: body parts are not placed at different positions on a big spritesheet. They all occupy the same 48x48 area and are painted over each other in a specific order.

Layer Stacking Order (Back to Front)

The compositor paints body parts in this exact order:

Order	Body Part	What It Contains
1st	Carapace	Back shell, main body shape & color
2nd	Legs	Walking appendages, lower body
3rd	Tail	Tail fan, rear propulsion
4th	Eyes	Eye stalks, eye shape & color
5th	Antennae	Sensory appendages, glow effects
6th	Claws	Front claws, weapons (always on top)

Parts drawn later cover parts drawn earlier. Claws are always visible on top. The carapace forms the background silhouette.

The Designer Workflow in Composite Mode

Step 1: Enable Composite Mode

Check the **Composite** checkbox in the toolbar. This: - Creates 6 default layers (one per body part) - Shows colored filter buttons for toggling body part visibility - Hides the single-part “Part” dropdown (not needed — you’re working on all parts) - Shows a “Comp” composite preview row

Step 2: Draw the Whole Lobster

Draw your lobster character on the canvas, using **different layers for different body parts**. Each layer is assigned to a body part via the colored dropdown in the layer panel.

For example: - Select the “**Cap Base**” layer → draw the shell/back - Select the “**Claw Base**” layer → draw the claws - Select the “**Eye Base**” layer → draw the eyes

You’re drawing everything on the **same 64x64 canvas**. The parts physically overlap — the carapace pixels and the claw pixels both exist in the same space, just on different layers.

Step 3: Use the Filter Bar to Focus

The colored buttons in the toolbar (Cap, Claw, Tail, Ant, Eye, Leg) let you hide/show body parts:

- Click “**Claw**” to hide all claw layers — now you can see what’s behind them
- Click it again to show claws
- This helps you see how parts overlap and ensure the stacking looks correct

Step 4: Check the Composite Preview

The “**Comp**” row in the preview bar shows 4 small canvases — one per evolution tier (Base, Evolved, Elite, Apex). These show the fully assembled lobster with all visible body parts composited in the correct stacking order.

Step 5: Add Tier Progression

Each body part can have multiple layers at different evolution tiers:

Tier	Name	Canvas Region	Purpose
T0	Base	Center ~42x42	Essential silhouette, compact form
T1	Evolved	Center ~52x52	More body detail, bulk
T2	Elite	Center ~58x58	Textures, patterns, class markings
T3	Apex	Full 64x64	Spikes, auras, elaborate flourishes

To add tier layers: click “+ **Layer**”, set the tier in the layer panel, and assign the body part. A typical setup might be:

```
Cap Base  (T0, Carapace)  ← shell silhouette
Cap Evol  (T1, Carapace)  ← shell detail
Cap Elite (T2, Carapace)  ← shell textures
Cap Apex  (T3, Carapace)  ← shell ornaments
Claw Base (T0, Claws)     ← claw silhouette
Claw Evol (T1, Claws)     ← claw detail
... etc for all 6 body parts
```

Step 6: Export

Use **File > Export All Parts JSON** to generate 6 separate template files — one per body part. Each file contains only the pixels from layers assigned to that body part, with proper tier tags, anchor points, and mutation zones.

You can also export a single part via **File > Export JSON (Ctrl+S)** — this exports all layers (regardless of body part assignment) as a single template, using the active layer’s body part.

How the Palette Role System Works

You don't paint with raw colors. Instead, you paint with **7 semantic roles** that automatically adapt to each of the 10 lobster classes:

Role	Name	Typical Use
0	Outline	Dark contour lines, silhouette edges
1	Primary Shadow	Darkest shading on main body color
2	Primary Base	Main body color (~50% of pixels)
3	Primary Highlight	Light areas, specular hits
4	Secondary Base	Accent areas, belly, joints
5	Secondary Highlight	Light areas on secondary color
6	Accent	Small pops of contrast (eyes, gems)

One template, 10 class renderings. A carapace painted with these roles automatically looks correct in Bulwark's steel blue, Ember's fiery orange, Abyss's dark green, etc.

Custom colors (roles 7+) are fixed across all class palettes — use them for class-agnostic details like legend effects.

Why Not a Bigger Canvas?

You might expect a larger canvas where body parts are drawn side-by-side and then sliced apart. That's not how Clawbada works. Here's why:

The game composites at runtime. When a player's lobster is rendered, the game: 1. Loads 6 separate body part templates (one per part) 2. Paints each one onto a 48x48 canvas at position (0, 0) 3. Stacks them back-to-front (Carapace first, Claws last)

Since every part is painted at the **same origin point**, they must be drawn in the same coordinate space. If the carapace shell occupies pixels (10,10) to (38,38), and a claw overlaps at (8,20) to (20,35), both exist on the same 48x48 grid.

The composite preview in the editor reproduces exactly what the game does — it layers all your body part layers in the correct order so you can see the final result while you work.

Anchor Points

Each body part has a reference anchor point that defines its “center of gravity” on the 48x48 canvas:

Body Part	Anchor (x, y)	Description
Carapace	(24, 20)	Center-back
Claws	(12, 28)	Front-center
Tail	(34, 36)	Back-low
Antennae	(18, 10)	Top-front
Eyes	(16, 18)	Front-high
Legs	(24, 38)	Bottom-center

Anchors ensure body parts align correctly during compositing. The editor auto-sets anchors based on the body part assignment.

Mutation Zones

Mutation zones are rectangular regions where the game's procedural system can modify pixels to create visual variants. Each lobster's DNA specifies a variant number (0-15), and the game applies small changes within these zones — adding pixels, removing them, shifting patterns, etc.

In composite mode, zones are global — they apply to whichever body part's pixels happen to be within the zone's rectangle. A zone drawn over the claw area only affects claw pixels because that's the only body part with pixels in that region.

Quick Reference: Keyboard Shortcuts

Shortcut	Action
Ctrl+S	Export single-part JSON
Ctrl+Z	Undo
Ctrl+Shift+Z	Redo
B	Pencil tool
E	Eraser tool
G	Fill tool
I	Eyedropper tool
L	Line tool
R	Rectangle tool
Middle-click drag	Pan canvas
Scroll wheel	Zoom in/out

Summary

1. **All 6 body parts share the same canvas** — they overlap, not side-by-side
2. **Draw on separate layers** — assign each layer to a body part
3. **The filter bar** lets you hide/show parts to focus on one area
4. **The composite preview** shows the assembled lobster in real-time
5. **Export All Parts** generates 6 JSON files automatically
6. **Paint with roles** (0-6) so one template works across all 10 classes
7. **Add tier layers** (T0-T3) for evolution visual progression